Assignment 3

Task 1: Independent Sample T-Test

🎯 Objective:

Compare CGPA between Male and Female students.

✅ SPSS Steps:

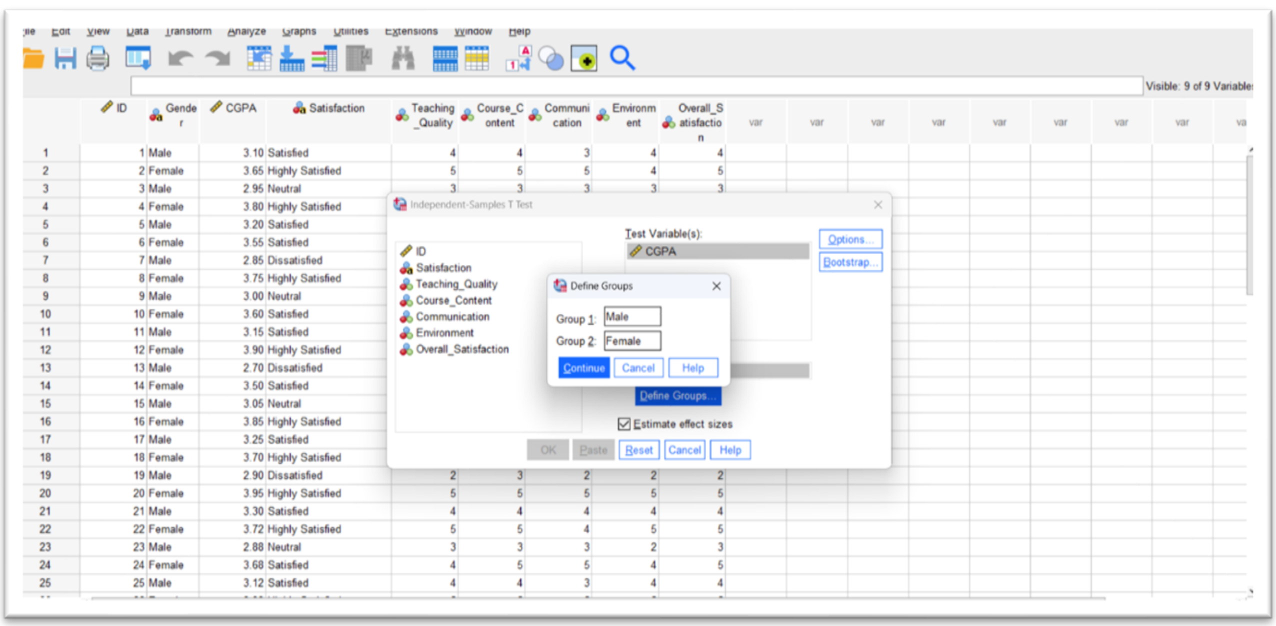
Analyze → Compare Means → Independent Sample T-Test

Test Variable:

- CGPA

Grouping Variable:

- Gender



Group Statistics					
	Gender	N	Mean	Std. Deviation	Std. Error Mean

CGPA	Male	20	3.0240	.17080	.03819
	Female	20	3.7380	.13173	.02946

Mean CGPA(Male): 3.0240

Mean CGPA(Female): 3.7380

Independent Samples Test										
		Levene's Test for Equality of Variances		t-test for Equality of Means						
		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
CGPA	Equal variances assumed	1.794	.188	-14.803	38	<.001	-.71400	.04823	-.81164	-.61636
	Equal variances not assumed			-14.803	35.697	<.001	-.71400	.04823	-.81185	-.61615

Interpretation

The Levene's Test significance value is 0.188, which is greater than 0.05. Therefore, the "Equal variances assumed" row is used for interpretation.

The Independent Sample T-Test shows that the p-value is < 0.001, indicating a statistically significant difference in CGPA between male and female students.

The mean difference is -0.714, which suggests that the average CGPA of the two groups is not equal. Therefore, it can be concluded that academic performance differs significantly based on gender.

Task 2: Chi-Square Test

🎯 Objective:

Check relationship between Gender and Satisfaction.

✅ SPSS Steps:

Analyze → Descriptive Statistics → Crosstabs

Row:

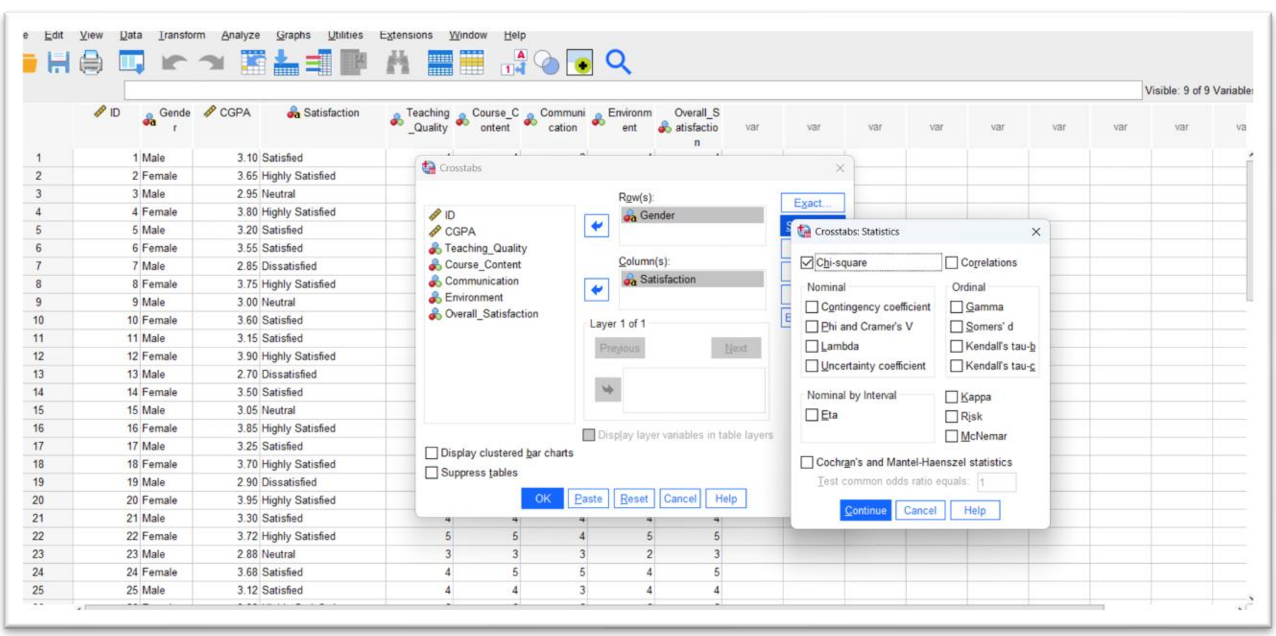
- Gender

Column:

- Satisfaction

Then:

- Statistics → Chi-square



Chi-Square Tests			
	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	26.286[a]	3	.000
Likelihood Ratio	36.330	3	.000
N of Valid Cases	40		
a. 4 cells (50.0%) have expected count less than 5. The minimum expected count is 2.50.			

Interpretation

The Pearson Chi-Square value is 26.286 with df = 3. The p-value is < 0.001, which is less than 0.05.

Therefore, the relationship between Gender and Satisfaction is statistically significant. This indicates that satisfaction levels vary significantly between male and female students.

Task 3: Reliability Test

🎯 Objective:

Check reliability of Likert Scale questions.

✅ Variables:

- Teaching_Quality
- Course_Content
- Communication
- Environment
- Overall_Satisfaction

✅ SPSS Steps:

Analyze → Scale → Reliability Analysis

Model = Alpha

Reliability Statistics	
Cronbach's Alpha	N of Items
.967	5

Interpretation Guide

Cronbach's Alpha	Reliability Level
< 0.60	Poor
0.60 – 0.69	Questionable
0.70 – 0.79	Acceptable
0.80 – 0.89	Good
≥ 0.90	Excellent

Interpretation

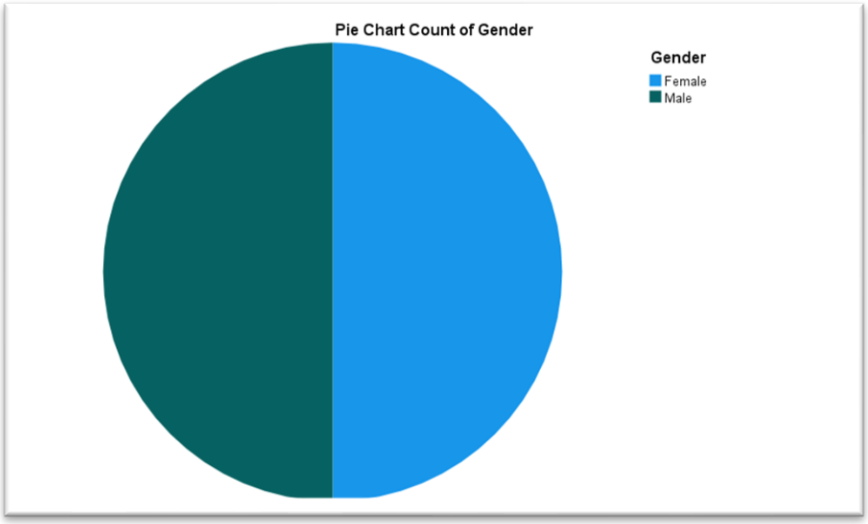
The Cronbach's Alpha value is 0.967 for 5 items, which indicates excellent internal consistency among the Likert scale questions.

Since the alpha value is greater than 0.90, the questionnaire is highly reliable, and the items consistently measure the same concept.

Task 4: Graph

Create:

- 1 Bar Chart
  OR
- 1 Pie Chart



Pie Chart Count of Gender
Gender
Female
Male